

AI — RAG, Fine-Tuning & Pre-training

1. RAG — Retrieval-Augmented Generation

Definition

RAG (Retrieval-Augmented Generation) is a technique where an existing **Large Language Model (LLM)** uses external knowledge sources to retrieve relevant information and generate an answer based on that information.

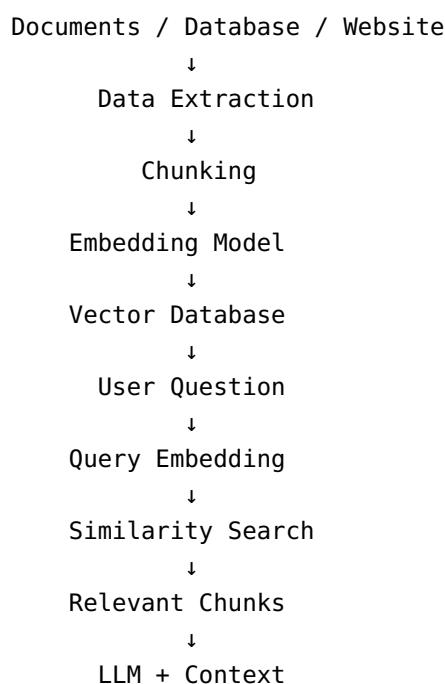
Simple-ah:

Existing AI model-a retrain pannaama, namma own data-la irundhu relevant information-a retrieve panni, adha LLM-ku context-ah kuduthu answer generate panna vaikkaradhu dhaan RAG.

Examples of External Data

- PDF
- Word documents
- Database
- Website
- API
- Company documents
- Knowledge base

RAG Pipeline



↓
Generated Answer

Important Technical Terms

Chunking

Large document-a smaller text portions-ah split pannradhu.

Example:

```
Large PDF
  ↓
Chunk 1
Chunk 2
Chunk 3
Chunk 4
```

Embedding

Text-a numerical vector representation-ah convert pannradhu.

Example:

```
"Python is a programming language"
  ↓
Embedding Model
  ↓
[0.21, -0.43, 0.87, ...]
```

Embedding model dhaan text-a vector-ah convert pannum.

Vector

Text-oda semantic meaning-a represent panna use pannra numerical representation.

Example:

```
Text
  ↓
[0.12, 0.45, -0.72, 0.31, ...]
```

Vector Database

Embeddings-a store panni, similar information-a quickly search panna use pannra database.

Examples:

- Pinecone
- Qdrant
- Weaviate
- Chroma
- Milvus
- pgvector

Important:

Vector Database text-a vector-ah convert pannaadhu.

Usually:

```
Text
↓
Embedding Model
↓
Vector
↓
Vector Database
```

Retrieval

User question-ku relevant information-a database-la irundhu search panni edukkardhu.

Example:

```
User:
"College closing time enna?"

↓

Vector Search

↓

Relevant Chunk:
"College working hours are 9 AM to 5 PM."
```

Augmentation

Retrieved information-a LLM-ku additional context-ah provide pannradhu.

```
graph TD
    A[User Question] -- "+" --> B[Retrieved Information]
    B -- "↓" --> C[LLM Context]
```

Generation

LLM retrieved context-a use panni final answer generate pannradhu.

Important Point

RAG hallucination-a **reduce** panna help pannum.

But:

RAG hallucination-a completely eliminate pannaadhu.

Wrong information retrieve aana, retrieval poor-ah irundha, or LLM context-a wrong-ah interpret panna, hallucination still occur aagalam.

2. Fine-Tuning

Definition

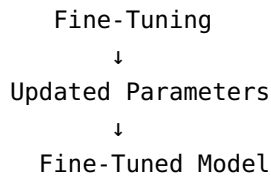
Fine-Tuning is the process of adapting an already pre-trained model using a task-specific dataset by updating the model's parameters/weights.

Simple-ah:

Already trained model-ku specific task, behavior, format, style-ku additional training kudukkaradhu dhaan Fine-Tuning.

Basic Process

```
graph TD
    A[Pre-trained LLM] -- "↓" --> B[Training Dataset]
    B -- "↓" --> C[Tokenization]
    C -- "↓" --> D[ ]
```



Example

Normal LLM:

User:
Explain this code.

Model:
General explanation.

Fine-tuned model:

User:
Explain this code.

Model:
Unga organization define pannina
specific format/style-la explanation.

Fine-Tuning Use Cases

Fine-Tuning can be used for:

- Specific response style
- Instruction following
- Classification
- Domain-specific tasks
- Structured output
- Specialized behavior
- Conversation style

Dataset

Fine-Tuning dataset necessarily Question-Answer pairs mattum irukka vendiya avasiyam illa.

Example:

```
{
  "instruction": "Explain REST API",
  "input": "",
```

```
"output": "REST API is..."
}
```

Or conversational format:

```
User: Explain REST API.
Assistant: REST API is...
```

LoRA

LoRA = Low-Rank Adaptation

LoRA is a **Parameter-Efficient Fine-Tuning (PEFT)** technique.

Full model-oda parameters ellathayum update pannaama, small set of additional trainable parameters use panni model-a adapt pannum.

Full Fine-Tuning vs LoRA

```
Full Fine-Tuning
→ Model weights update
→ More GPU memory
→ More expensive

LoRA
→ Base model mostly frozen
→ Small adapter parameters trained
→ Less memory
→ Cheaper
```

3. Pre-training

Definition

Pre-training is the large-scale process of training a neural network from its initial state using massive datasets so that it learns general language and world patterns.

Simple-ah:

Oru AI model-a basic level-la irundhu train panni, general knowledge and language patterns learn panna vaikkaradhu dhaan Pre-training.

Basic Process

```
Massive Dataset
  ↓
Data Cleaning
  ↓
Tokenization
  ↓
Tokens
  ↓
Transformer Model
  ↓
GPU / TPU
  ↓
Training
  ↓
Pre-trained LLM
```

Example

Model-ku:

"The capital of France is ____"

nu kudutha:

Paris

nu predict panna kathukkum.

Large-scale training-la billions/trillions of tokens use pannitu language patterns, relationships, reasoning-related capabilities etc. learn pannum.

RAG vs Fine-Tuning vs Pre-training

Feature	RAG	Fine-Tuning	Pre-training
Existing model use?	Yes	Yes	No
Model weights change?	No	Yes	Yes
External data at runtime?	Yes	Usually No	No
Main purpose	Knowledge retrieval	Behavior / task adaptation	General model learning

Feature	RAG	Fine-Tuning	Pre-training
Training required?	No	Yes	Yes
Cost	Low	Medium	Extremely High
Difficulty	Easy	Medium / Hard	Very Hard

Simple Real-World Analogy

Imagine **LLM = Engineer**.

Pre-training

Engineer-a school + college-la zero-la irundhu educate pannradhu.

```

Zero knowledge
  ↓
Education
  ↓
General knowledge

```

Fine-Tuning

Already knowledgeable engineer-ku specific company/process/technology training kudukkaradhu.

```

Existing Engineer
  ↓
Specialized Training
  ↓
Specialized Engineer

```

RAG

Engineer-ku retraining kudukkaama, office documents/manuals kuduthu, question varumbothu relevant document-a paathu answer panna solradhu.

```

Engineer
  +
Company Documents
  ↓
Search relevant document

```

↓
Answer

Most Important Difference

RAG ≠ Training

RAG use pannumbothu LLM usually learn/retrain aagathu.

Example:

```
Gemini
+
Your College Database
+
Vector Database
↓
RAG Application
```

User question varumbothu:

```
User Question
↓
Convert question to embedding
↓
Similarity Search
↓
Retrieve relevant data
↓
Add data to context
↓
Gemini
↓
Answer
```

Tomorrow database-la new information add pannina:

```
Old:
Course A → HOD = Kumar

New:
Course A → HOD = Ravi
```

Database / vector index update pannina podhum.

Gemini-a retrain panna thevai illa.

Important Architecture

RAG understand panna indha sequence romba important:

```
LLM
↓
Token
↓
Transformer
↓
Embedding
↓
Vector
↓
Vector Database
↓
Similarity Search
↓
Retrieval
↓
Context
↓
LLM
↓
Answer
```

Key Terminology — Quick Revision

LLM → Large Language Model.

Token → Text-ai model process panna use pannra basic units.

Embedding → Text/data-ai numerical vector representation-ah convert pannradhu.

Vector → Data-oda semantic representation in numerical form.

Vector Database → Vectors-a store, index, search panna use pannra database.

Chunking → Large documents-a smaller text sections-ah split pannradhu.

Retrieval → Query-ku relevant information-a search panni retrieve pannradhu.

Augmentation → Retrieved information-a LLM context-kulla add pannradhu.

Generation → LLM final response generate pannradhu.

RAG → Retrieval + Context + LLM Generation.

Fine-Tuning → Existing model-a task/behavior-specific dataset-la additional training pannradhu.

LoRA → Parameter-Efficient Fine-Tuning technique.

Pre-training → Massive datasets use panni model-ai general capabilities learn panna train pannradhu.

Inference → Trained model-ai use panni actual user input-ku output generate pannradhu.

One-Line Memory Trick

PRE-TRAINING

= Brain create & general knowledge learn

FINE-TUNING

= Existing brain-ku specialization training

RAG

= Brain-ku external reference book provide pannradhu